Q1 (a)

$$L(\lambda) = \prod_{i=1}^{n} f(x_i) = \prod_{i=1}^{n} \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$$

$$L(\lambda) = (e^{-\lambda} \lambda^{x_1}) \cdot (e^{-\lambda} \lambda^{x_2}) \cdots (e^{-\lambda} \lambda^{x_n})$$

$$L(\lambda) = e^{-n\lambda} \lambda^{\sum_{i=1}^{n} x_i} \prod_{i=1}^{n} \frac{1}{x_i!}$$

$$\ln L(\lambda) = \ln\left(e^{-n\lambda} \lambda^{\sum_{i=1}^{n} x_i} \prod_{i=1}^{n} \frac{1}{x_i!} \right)$$

$$\ln L(\lambda) = -n\lambda + \left(\sum_{i=1}^{n} x_i \right) \ln\lambda - \sum_{i=1}^{n} \ln(x_i!)$$

$$\frac{d}{d\lambda} \ln L(\lambda) = \frac{-n + \sum_{i=1}^{n} x_i}{\lambda}$$

$$\frac{-n + \sum_{i=1}^{n} x_i}{\lambda} = 0$$

$$\hat{\lambda} = \frac{\sum_{i=1}^{n} x_i}{n}$$

G1(b)

$$L(\mu,\sigma) = \prod_{i=1}^{n} f(x_i) = \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}}$$

$$L(\mu,\sigma) = \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^n e^{-\sum_{i=1}^{n}\frac{(x_i-\mu)^2}{2\sigma^2}}$$

$$\ln L(\mu,\sigma) = \ln\left(\left(\frac{1}{\sigma\sqrt{2\pi}}\right)^n e^{-\sum_{i=1}^{n}\frac{(x_i-\mu)^2}{2\sigma^2}}\right)$$

$$\ln L(\mu,\sigma) = -n\ln(\sigma\sqrt{2\pi}) - \sum_{i=1}^{n}\frac{(x_i-\mu)^2}{2\sigma^2}$$

$$\frac{\partial}{\partial\mu}\ln L(\mu,\sigma) = \sum_{i=1}^{n}\frac{2(x_i-\mu)}{2\sigma^2}$$

$$\frac{\partial}{\partial\mu}\ln L(\mu,\sigma) = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu)$$

$$\frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu) = 0$$

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\hat{\mu} = \bar{x}$$

$$\frac{\partial \ln L(\mu,\sigma)}{\partial \sigma} = \frac{-n}{\sigma} + \frac{1}{\sigma^3}\sum_{i=1}^{n}(x_i-\mu)^2$$

$$\frac{-n}{\sigma} + \frac{1}{\sigma^3}\sum_{i=1}^{n}(x_i-\mu)^2 = 0$$

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2$$

$$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2}$$

Q2

i)

→ For $\hat{\mu}_1 = \frac{X_1+X_2+X_3}{3}$:

$$E(\hat{\mu}_1) = \frac{E(X_1)+E(X_2)+E(X_3)}{3} = \frac{\mu+\mu+\mu}{3} = \mu$$

$$\text{Bias}(\hat{\mu}_2) = E(\hat{\mu}_1) - \mu = \mu - \mu = 0$$

$\hat{\mu}_1$ is unbiased

→ For $\hat{\mu}_2 = \frac{X_1+X_2}{4} + \frac{X_3}{5}$:

$$E(\hat{\mu}_2) = \frac{E(X_2)+E(X_2)}{4} + \frac{E(X_3)}{5}$$

$$E(\hat{\mu}_2) = \frac{\mu}{4} + \frac{\mu}{5}$$

$$E(\hat{\mu}_2) = \frac{2\mu}{4} + \frac{\mu}{5}$$

$$E(\hat{\mu}_2) = \frac{\mu}{2} + \frac{\mu}{5} = \frac{7\mu}{10}$$

$$\text{Bias}(\hat{\mu}_2) = E(\hat{\mu}_2) - \mu = \frac{7\mu}{10} - \mu = \frac{-3\mu}{10}$$

$\hat{\mu}_2$ is biased

→ For $\hat{\mu}_3 = \frac{X_1}{6} + \frac{X_2}{4} + \frac{X_3}{4} + 2$:

$$E(\hat{\mu}_3) = \frac{E(X_1)}{6} + \frac{E(X_2)}{4} + \frac{E(X_3)}{4} + 2$$

$$E(\hat{\mu}_3) = \frac{\mu}{6} + \frac{\mu}{4} + \frac{\mu}{4} + 2$$

$$E(\hat{\mu}_3) = \frac{2\mu}{6} + \frac{\mu}{4} + 2$$

$$E(\hat{\mu}_3) = \frac{\mu}{3} + \frac{\mu}{4} + 2 = \frac{7\mu}{12} + 2$$

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$$\text{Bias}(\hat{\mu}_3) = E(\hat{\mu}_3) - \mu = \frac{7\mu}{12} + 2 - \mu = \frac{-5\mu}{12} + 2$$

$\hat{\mu}_3$ is biased

ii)

→ For $\hat{\mu}_1 = \frac{X_1 + X_2 + X_3}{3}$:

$$\text{Var}(\hat{\mu}_1) = \frac{1^2 \cdot \text{Var}(X_1) + 1^2 \cdot \text{Var}(X_2) + 1^2 \cdot \text{Var}(X_3)}{9}$$

$$\text{Var}(\hat{\mu}_1) = \frac{7 + 13 + 20}{9} = \frac{40}{9} \approx 4.44$$

→ For $\hat{\mu}_2 = \frac{X_1 + X_2}{4} + \frac{X_3}{5}$:

$$\text{Var}(\hat{\mu}_2) = \frac{1^2 \cdot \text{Var}(X_1) + 1^2 \cdot \text{Var}(X_2)}{16} + \frac{1^2 \cdot \text{Var}(X_3)}{25}$$

$$\text{Var}(\hat{\mu}_2) = \frac{7 + 13}{16} + \frac{20}{25} = \frac{20}{16} + \frac{20}{25}$$

$$\text{Var}(\hat{\mu}_2) = 1.25 + 0.8 = 2.05$$

→ For $\hat{\mu}_3 = \frac{X_1 + X_2}{6} + \frac{X_3}{4} + 2$:

$$\text{Var}(\hat{\mu}_3) = \frac{1^2 \cdot \text{Var}(X_1) + 1^2 \cdot \text{Var}(X_2)}{36} + \frac{1^2 \cdot \text{Var}(X_3)}{16}$$

$$\text{Var}(\hat{\mu}_3) = \frac{7 + 13}{36} + \frac{20}{16} = \frac{20}{36} + \frac{20}{16}$$

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$$Var(\hat{\mu}_3) = 0.5556 + 1.25 = 1.8056$$

iii)

$$MSE(\hat{\mu}) = Var(\hat{\mu}) + Bias^2(\hat{\mu})$$

→ For $\hat{\mu}_1$:

$$MSE(\hat{\mu}_1) = 4.44 + 0^2 = 4.44$$

→ For $\hat{\mu}_2$:

$$MSE(\hat{\mu}_2) = 2.05 + \left(\frac{-3\mu}{10}\right)^2 = 2.05 + \left(\frac{9\mu^2}{100}\right)$$

→ For $\hat{\mu}_3$:

$$MSE(\hat{\mu}_3) = 1.8056 + \left(\frac{-5\mu}{12} + 2\right)^2$$

Q3

Given:

$\mu_0 = 1120$ hours $\sigma = 125$ hours $n = 100$

$\bar{x} = 1070$ hours $\alpha = 5\%$ or 0.05.

Hypotheses:

- **Null Hypotheses (H_0):** The mean lifetime has not changed i.e. $\mu = 1120$ hours.
- **Alternative Hypotheses (H_a):** The mean lifetime has changed i.e. $\mu \neq 1120$ hours (two-tailed test).

Test Statistic:

z-test for mean

$$z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

$$z = \frac{1070 - 1120}{\frac{125}{\sqrt{100}}} = \frac{-50}{\frac{125}{10}} = \frac{-50}{12.5} = -4$$

(a) Critical Value Approach:

1. For a two-tailed test at $\alpha = 0.05$, we need to find the critical values corresponding to $\alpha/2 = 0.025$ in each tail

From z-tables, the critical z-value for a two tailed test at $\alpha = 0.05$ is approximately ± 1.9

2. Compare the calculated z-value to the critical values:
   - if $|z| > 1.96$, we reject the null hypothesis
   - if $|z| \leq 1.96$, we fail to reject null hypothesis

Since $|z| = 4 > 1.96$, we reject the null hypothesis

(b) P-value approach

1. Find the p value corresponding to the z-score of $z = -4$ using a standard normal distribution table or a calculator
   - The p value for $z = -4$ is approx 0.00003

2. Compare the p value to the significance level $\alpha = 0.05$

- If the p value is less than α, we reject the null hypothesis.
- If the p value is greater than α, we fail to reject the null hypothesis

Since $0.00003 < 0.05$, we reject the null hypothesis

(c) Confidence Interval Approach:

1. Construct a 95% confidence interval for the population mean. The formula for the confidence interval is

$$= \bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

$$= 1070 \pm 1.96 \cdot \frac{125}{\sqrt{100}} = 1070 \pm 1.96 \cdot 12.5$$

$$= 1070 \pm 24.5$$

Thus the 95% confidence interval is

$$(1045.5, 1094.5)$$

2. Compare the null hypotheses mean $\mu_0 = 1120$ to the confidence interval

- If μ_0 lies within the confidence interval, we fail to reject the null hypotheses.
- If μ_0 lies outside the confidence interval, we reject the null hypotheses.

Since 1120 is outside the interval (1045.5, 1094.5), we reject the null hypothesis.

→ Thus, based on the data and 5% significance level, we conclude that the mean lifetime of the bulbs has changed.

Q4

Hypotheses:

- **Null Hypothesis (H_0):** The mean cholesterol level is 3.1 i.e. $\mu = 3.1$
- **Alternative Hypothesis (H_a):** The mean cholesterol level is not 3.1 i.e. $\mu \neq 3.1$ (two-tailed test)

Test Statistic:

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

Sample Mean ($\bar{x}$):

$$\bar{x} = \frac{3.1 + 2.8 + 1.5 + 1.7 + 2.4 + 1.9 + 3.3 + 1.6}{8}$$

$$\bar{x} = \frac{16.3}{8} = 2.0375$$

Sample Standard Deviation (s):

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

Calculate squared differences from mean:

$(3.1 - 2.0375)^2 = 1.1264$
$(2.8 - 2.0375)^2 = 0.5856$
$(1.5 - 2.0375)^2 = 0.2856$
$(1.7 - 2.0375)^2 = 0.1139$
$(2.4 - 2.0375)^2 = 0.1314$
$(1.9 - 2.0375)^2 = 0.0190$
$(3.3 - 2.0375)^2 = 1.5876$
$(1.6 - 2.0375)^2 = 0.1901$

Sum of squared differences = 4.0496

$$s^2 = \frac{4.0496}{7} = 0.5785$$

$$s = \sqrt{0.5785} \approx 0.759$$

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$$t = \frac{2.0375 - 3.1}{\frac{0.759}{\sqrt{8}}}$$

$$t = \frac{-1.0625}{\frac{0.75}{2.828}}$$

$$t = \frac{-1.0625}{0.2684}$$

$$t \approx -3.95$$

(a) Critical Value Approach

1. For a two-failed test $\alpha = 0.05$, the degree of freedom is $n-1 = 8-1 = 7$.
   From the t-distribution table, the critical t value for $\alpha/2 = 0.025$ and 7 degrees of freedom is approx ± 2.365
2. Comparing calculated t-value to critical value
   Since $|t| = 3.95 > 2.365$, we reject the null hypothesis.

(b) P-value approach

1. Finding the p value corresponding to $t = -3.95$ with 7 degrees of freedom which is approx 0.004
2. Comparing p-value to significance level $\alpha = 0.05$
   Since $p = 0.004 < 0.05$, we reject null hypothesis

(c) Confidence Interval Approach

$$: \bar{x} \pm t_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

$$: 2.0375 + 2.365, \frac{0.759}{\sqrt{8}}$$

$$= 2.0375 \pm 2.365 \cdot 0.2684$$

$$= 2.0375 \pm 0.635$$

Thus, the 95% confidence interval is
(1.4025, 2.6725)

Comparing mean $\mu_0 = 3.1$ to confidence interval
Since 3.1 is outside the interval (1.4025, 2.6725)
we reject the null hypothesis

→ Thus based on the data and at the 5% significance level, we conclude that the sample has not been drawn from a distribution with a mean cholestrol level of 3.1

G5

Hypotheses:

- **Null Hypothesis:** The mean rainfall in region A is not less than that of Region B i.e. $\mu_A \geq \mu_B$
- **Alternative Hypothesis:** The mean rainfall in Region A is less than that of Region B i.e. $\mu_A < \mu_B$ (this is a left-tailed test)

Test Statistic:

z-test for the difference of means

$$z = \frac{(\bar{x}_A - \bar{x}_B)}{\sqrt{\frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}}}$$

$$z = \frac{(700 - 725)}{\sqrt{\frac{66^2}{60} + \frac{66^2}{60}}} = \frac{-25}{\sqrt{\frac{4356}{60} + \frac{4356}{60}}}$$

$$z = \frac{-25}{\sqrt{145.1 + 145.2}} = \frac{-25}{\sqrt{290.4}}$$

$$z = \frac{-25}{17.03} \approx -1.47$$

(a) Critical Value Approach

The critical z-value for $\alpha = 0.01$ is approximately $z_{0.01} = -2.33$

Since $z = -2.47$ is greater than -2.33, we fail to reject the null hypothesis

(b) P-value Approach

The p-value for $z = -2.47$ is approx 0.0708

Since $0.0708 > 0.01$, we fail to reject the null hypothesis.

(c) Confidence Interval Approach

$$= (\bar{x}_A - \bar{x}_B) \pm z_{\alpha} \cdot \sqrt{\frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}}$$

$$= -25 \pm 2.33 \cdot \sqrt{\frac{66^2}{60} + \frac{66^2}{60}}$$

$$= -25 + 2.33 \cdot 17.03$$

$$= -25 \pm 39.7$$

Thus, the 95% confidence interval is
(-64.7, 14.7)

Since the confidence interval (-64.7, 14.7) contains zero, we fail to reject the null hypothesis

→ Thus based on the data and at the 1% significance level we conclude that the sample data does not support the researcher's claim.

G6

Hypotheses:

- **Null Hypothesis:** The mean daily expenses for sales staff are less than or equal to the mean daily expenses for audit staff
- **Alternative Hypothesis:** The mean daily expenses for sales staff are greater than those for audit staff

Sample Mean & Standard Deviation

$$\bar{x}_2 = \frac{131 + 135 + 146 + 165 + 136 + 142}{6}$$

$$\bar{x}_1 = \frac{855}{6} = 142.5$$

$$s_2 = \sqrt{\frac{\sum_{i=2}^{n} (x_i - \bar{x}_2)^2}{n_2 - 1}}$$

$$s_1 = \sqrt{\frac{749.5}{5}} \approx \sqrt{149.9} \approx 12.25$$

$$s_2 = \sqrt{\frac{1476.58}{6}} \approx \sqrt{246.1} \approx 15.7$$

Test Statistic: